

GMCL1 Controls 53BP1 Stability and Modulates Paclitaxel Sensitivity in Cancer

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eLife Assessment

This study identifies 53BP1 as an interaction partner of GMCL1 (a likely CUL3 substrate receptor). The study seeks to link this finding to regulation of the mitotic surveillance pathway and paclitaxel resistance in cancer. The evidence for these claims is currently **inadequate**; concerns include the use of cell lines that have been reported to lack the mitotic surveillance pathway, insufficient consideration of paclitaxel mechanisms of action, and an overinterpretation of correlative results.

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Abstract

The Mitotic Surveillance Pathway (MSP) monitors the duration of M-phase. Prolonged mitosis, caused by spindle attachment defects or microtubule-targeting drugs such as the taxane paclitaxel, induces the formation of the ternary “mitotic stopwatch” complex consisting of 53BP1, USP28, and p53. This event protects p53 from degradation, resulting in cell cycle arrest or apoptosis in daughter cells. In paclitaxel-resistant cancers, cells bypass the MSP, enabling unchecked proliferation and survival, although the underlying mechanisms remain unknown. Here, we demonstrate that 53BP1 physically interacts with GMCL1 but not its paralog, GMCL2, and we mapped the interaction regions on both proteins. CRL3^{GMCL1} functions as a ubiquitin ligase that targets 53BP1 for degradation during M phase, impacting p53 levels in daughter cells. High GMCL1 expression significantly correlates with resistance to paclitaxel in cancer cell lines with wild-type p53, including endometrial, breast, and upper aerodigestive tract cancer cells. Loss of GMCL1 restores paclitaxel sensitivity in p53 expressing cells but not in p53 deficient cells. We propose that in cancers with high GMCL1

levels, the CRL3^{GMCL1}-mediated degradation of 53BP1 prevents the formation of the mitotic stopwatch complex, leading to p53 degradation and sustained proliferation. Finally, our results indicate that GMCL1 inhibition represents a novel strategy to restore taxane sensitivity in resistant cancers.

Introduction

Mitosis is orchestrated by several intracellular signaling pathways to ensure proper cell division while maintaining genomic integrity. Errors during cell division, including chromosome mis-segregation or spindle defects, could lead to changes in chromosome content, producing aneuploid or polyploid progeny cells, which could be detrimental during development or lead to oncogenesis^{1,2}. Therefore, cells have evolved quality control mechanisms to ensure proper cell division during M phase. One such surveillance mechanism is known as the Mitotic Surveillance Pathway (MSP), which senses prolonged mitosis as an indicator of mitotic defects³.

If a threshold mitotic duration is surpassed, the MSP induces cell cycle arrest in the daughter cells in a p53-dependent manner^{4–6}. During abnormally prolonged mitosis^{7–10}, the MSP is triggered by the formation of a ternary “mitotic stopwatch” complex consisting of the deubiquitinase USP28 (ubiquitin-specific protease 28), 53BP1 (p53-binding protein 1), which acts as a substrate recognition subunit, and p53. During delayed mitosis, formation of the mitotic stopwatch complex allows USP28 to deubiquitinate and subsequently protect p53 from MDM2-mediated degradation. The 53BP1-USP28-p53 complex accumulates and persists in the G1 phase of daughter cells, where it promotes cell cycle arrest via p53-mediated upregulation of p21 expression.

Notably, proper mitotic arrest plays a crucial role in the mechanism of microtubule-targeting therapies, such as taxanes (e.g., paclitaxel and docetaxel), which disrupt spindle formation and chromosomal segregation. Recent studies have shown that pharmacological interventions targeting M phase, even without resulting in complete mitotic arrest, can induce controlled cell death¹¹. For example, low dose paclitaxel treatments results in increased p21 expression and apoptosis in daughter cells¹². However, there are cancers that display resistance or insensitivity to treatment with taxanes, including metastatic breast cancer, ovarian cancer, and non-small cell lung cancer^{13–15}. Therefore, there is an urgent need to elucidate the strategies of paclitaxel resistance in such cancers.

Human germ cell-less protein-like 1 (GMCL1) is a putative substrate receptor of one of many CUL3-RING ubiquitin ligases (CRL3s). However, to date, GMCL1's biological role and its substrates have remained uncharacterized. *GMCL1* received its namesake from its *Drosophila melanogaster* homolog, *GCL* (Germ Cell-Less), which plays essential roles in early embryonic development and germ cell determination^{16,17}. *GMCL1*'s role in germ cell development appears to have been evolutionarily conserved as loss of *GMCL1* expression in mice has been shown to cause defects in meiosis and spermiogenesis¹⁸, and altered *GMCL1* expression was functionally associated with human asthenozoospermia¹⁹. Although *GMCL1* homologs have been primarily associated with germ cell biology, several databases (e.g., GTEx²⁰ and GENT2²¹) indicate that *GMCL1* is expressed in somatic cells as well, suggesting broader biological functions beyond germ cell development. In contrast, the GMCL1 paralog GMCL2 is specifically expressed in germ cells.

Here, we present studies that shed light on the role of GMCL1 in somatic cells. We demonstrate that, like its *D. melanogaster* counterpart, GMCL1 acts as a CRL3 substrate receptor. We also identify 53BP1 as a *bona fide* substrate of CRL3^{GMCL1}, demonstrating that its levels are regulated by GMCL1 during M phase. Reduction of mitotic 53BP1 by GMCL1 prevents the formation of the

USP28-p53-53BP1 mitotic stopwatch complex and reduces p53 transmission to daughter cells. We therefore propose that GMCL1 inhibition, in combination with paclitaxel therapy, offers a promising strategy to overcome taxane resistance.

Results

Identification of 53BP1 as an interactor of CRL3^{GMCL1}

We have shown that the GMCL1 ortholog in *D. melanogaster*, GCL, is a substrate recognition subunit of a CRL3 complex that is active specifically in mitosis²². So, we predicted the human GMCL1 to also behave as a CRL3 substrate receptor. GMCL1 contains a BTB (Broad-Complex, Tramtrack, and Bric-a-brac) and a BACK (BTB and C-terminal Kelch) domain (**Supplementary Figure 1A**), consistent with other CRL3 receptors. On its C-terminus, GMCL1 also contains a GCL domain (residues 379-515), which is distinct from the Kelch domains commonly used by other CUL3 substrate receptors since it is predicted to form a β -sandwich characterized by two opposing anti-parallel β -sheets, each made up of four β -strands^{22,23} (**Supplementary Figure 1A**). A Dali search²⁴ of GMCL1's C-terminal domain reveals that it has some structural homology to the MATH (meprin and TRAF homology) domain used by another CRL3 substrate receptor, SPOP²⁵, suggesting that the predicted structure of GMCL1's GCL domain could also potentially act as a protein-protein interaction motif to recruit substrates.

To investigate the role of GMCL1 in somatic cells, we used immunoprecipitation followed by mass spectrometry (IP-MS) to identify binding partners of GMCL1. Proteomics studies were performed by expressing and purifying the following FLAG-tagged proteins: (i) wild-type GMCL1 (GMCL1 WT), (ii) GMCL1 E142K (GMCL1 EK), which carries a mutation in the BTB domain that is predicted to disrupt its interaction with CUL3, and (iii) GMCL1 BTB/BACK-only (GMCL1 BBO) that lacks the GCL domain (**Figure 1A-C**). IP-MS analysis identified 1,765 potential binding partners that specifically interact with GMCL1 via its C-terminal domain. Using SAINT scores > 0.70 and FDR < 5%, this list was refined to 9 proteins that showed significant interaction with GMCL1 WT and GMCL1 EK, but not with GMCL1 BBO, with 53BP1 being the one of most enriched protein (**Figure 1B**, **Supplementary Table 1**). This selective enrichment suggests that the C-terminal, "MATH-like" GCL domain of GMCL1 is critical for its interaction with binding partners, including 53BP1. The interaction between GMCL1 and endogenous 53BP1 and CUL3 was validated using immunoprecipitations (IPs) followed by immunoblotting (**Figure 1C,D**).

To further study the direct interaction between GMCL1 and 53BP1, we used AlphaFold 3²⁶ to predict their binding interface and identify potential critical residues at the recognition interface between the substrate receptor and 53BP1 to see if mutating these residues could abolish their interaction. Consistent with our initial IP-MS experiment, the AlphaFold 3 model predicted that the C-terminal domain of GMCL1 would interact with 53BP1. Based on the GMCL1-53BP1 complex structure prediction, we identified Arg 433, which appears to be a solvent-exposed residue in GMCL1's GCL domain that could interact with 53BP1 without impeding GMCL1's binding to with CUL3. Thus, we generated the GMCL1 R433A (GMCL1 RA) point mutant and tested its binding to 53BP1 using IP. As predicted, compared to WT GMCL1, the R433A mutation completely abolished the binding of GMCL1 to 53BP1, but did not impact GMCL1's binding to CUL3 (**Figure 1D**).

To determine which region of 53BP1 mediates its binding to GMCL1, we mapped the predicted GMCL1-binding site on 53BP1 and identified a conserved IEDI amino acid sequence within the Minimal Focus Forming region (MFF) of 53BP1²⁷ (**Supplementary Figure 1B-E**). Through IPs, we demonstrate that a 53BP1 mutant either lacking the MFF region or containing the IEDI-to-AAAA mutation within this region lost its interaction with CRL3^{GMCL1}, suggesting that this conserved IEDI sequence within 53BP1's MFF forms a critical degron recognized by GMCL1 (**Figure 1E**).

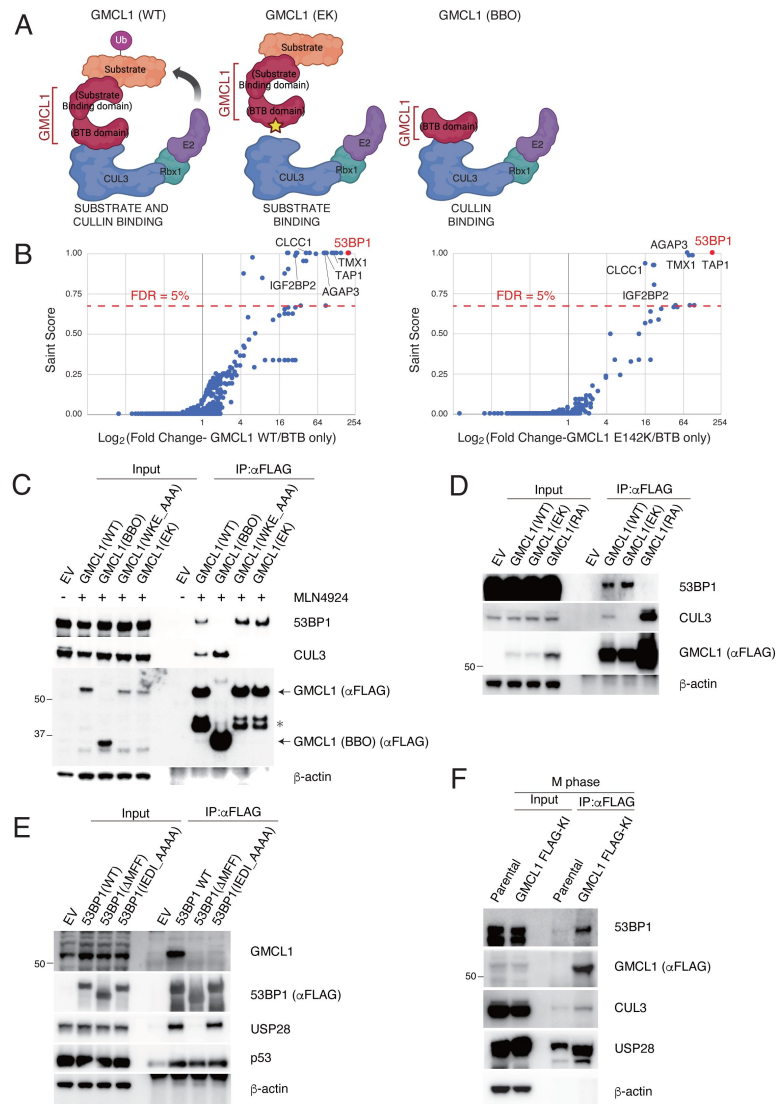


Figure 1.

Identification of 53BP1 as a GMCL1 interactor

A) Schematics for the immunoprecipitation-mass spectrometry (IP-MS) workflow using wild-type GMCL1 (GMCL1 WT) and mutants (GMCL1 EK and GMCL1 BBO). Color coding: Red, GMCL1; orange, putative substrates/interacting partners; blue, CUL3; green, RBX1; purple, E2 ubiquitin-conjugating enzyme.

B) HEK293T cells were transfected with FLAG-GMCL1 WT, FLAG-GMCL1 EK, or FLAG-GMCL1 BBO. After 24 hours, FLAG-tagged proteins were immunoprecipitated and analyzed by MS/MS. Left panel: proteins enriched with GMCL1 WT vs. BBO; right panel: proteins enriched with GMCL1 EK vs. BBO. Significant interactors were identified using SAINT scores > 0.70 and FDR < 5%.

C) HEK293T cells transfected with empty vector (EV), FLAG-GMCL1 WT, FLAG-GMCL1 BBO, FLAG-GMCL1 WKE_AAA (broadly disrupts the binding to CUL3) and FLAG-GMCL1 EK were treated with MLN4924 (3h). 53BP1 and CUL3 were immunoprecipitated with FLAG beads and analyzed by western blot. Asterisk indicates non-specific bands.

D) HEK293T cells were transfected with EV, FLAG-GMCL1 WT, FLAG-GMCL1 EK, or FLAG-GMCL1 RA. FLAG immunoprecipitations were probed for 53BP1 and CUL3.

E) HEK293T cells were transfected with EV, FLAG-53BP1 WT, FLAG-53BP1 ΔMFF and FLAG-53BP1 IEDI_AAAA. After MLN4924 treatment (3h), GMCL1 was immunoprecipitated and immunoblotted.

F) M phase-synchronized GMCL1 FLAG knock-in HCT116 cells were collected. GMCL1 was immunoprecipitated using FLAG-beads and analyzed by immunoblotting.

To examine the interaction between endogenous GMCL1 and endogenous 53BP1, we used CRISPR-Cas9 to knock in a FLAG tag at the C-terminus of *GMCL1*. Immunoprecipitation experiments confirmed that endogenous GMCL1 interacts with both endogenous CUL3 and 53BP1 (**Figure 1F** [↗](#)).

Finally, we sought to determine whether GMCL2, a GMCL1 paralog, is also able to interact with 53BP1. Immunoprecipitation of FLAG-tagged GMCL1 or GMCL2 from HEK293T cells revealed that while GMCL1 binds to 53BP1, GMCL2 does not (**Supplementary Figure 1F** [↗](#)), suggesting that GMCL1 and GMCL2 have distinct functions.

Overall, our results suggest that GMCL1 is a *bona fide* CRL3 substrate receptor and interacts specifically with 53BP1.

GMCL1 loss leads to 53BP1 accumulations under prolonged M phase

To investigate whether GMCL1 regulates 53BP1 stability, we generated *GMCL1* knock-out (KO) cells using CRISPR Cas-9²⁸ [↗](#) and compared 53BP1 levels between *GMCL1* WT and two *GMCL1* KO clones. We found that 53BP1 levels were significantly increased in our *GMCL1* KO cells during M phase. While the whole cell extracts (WCE) showed modest differences in GMCL1 levels between the *GMCL1* WT and KO clones, our fractionation experiments revealed that 53BP1 mainly accumulated in the chromatin-bound fraction of *GMCL1* KO cells (**Figure 2A** [↗](#)).

To further explore the role of GMCL1 on 53BP1 stability, we stably reconstituted U2OS *GMCL1* KO cells with either GMCL1 WT or binding mutants (i.e., GMCL1 EK and GMCL1 RA). Notably, the accumulation of 53BP1 in *GMCL1* KO cells was rescued only upon re-expression of GMCL1 WT. In contrast, 53BP1 levels remained high in *GMCL1* KO cells expressing either GMCL1 EK or GMCL1 RA (**Figure 2B** [↗](#)), emphasizing the importance of GMCL1's ability to bind both CUL3 (through the E142 residue) and 53BP1 (through the R433 residue) to regulate 53BP1 levels. Furthermore, GMCL1 WT re-expression in *GMCL1* KO cells led to decreased mRNA levels of p21 and NOXA during mitosis, whereas the GMCL1 EK and GMCL1 RA mutants failed to suppress these transcripts (**Supplementary Figure 2A** [↗](#)).

Collectively, these findings indicate that GMCL1 modulates 53BP1 availability through interactions with the CRL3^{GMCL1} [↗](#) complex, playing a critical role in the regulation of 53BP1 levels in M phase.

During mitosis, 53BP1 has been implicated in monitoring centrosome integrity and mitotic progression *via* the Mitotic Stopwatch Pathway (MSP). A recent study demonstrated that cells experiencing abnormally prolonged mitosis transmit this information to daughter cells through a ternary complex containing 53BP1, USP28, and p53 called the “mitotic stopwatch” complex⁷ [↗](#).

To investigate whether GMCL1 regulates this complex, we analyzed GMCL1 levels in post-mitotic daughter cells seven hours after release from prolonged nocodazole arrest. *GMCL1* KO daughter cells, reconstituted with GMCL1 WT, successfully re-entered the cell cycle, exhibiting low levels of 53BP1, p53, and p21, along with reduced expression of apoptosis-related genes (**Supplementary Figure 2B,C** [↗](#)). In contrast, cells reconstituted with either the E142K or R433A mutant displayed persistently elevated levels of 53BP1, p53, and p21, accompanied by increased expression of apoptosis-related genes (**Supplementary Figure 2B,C** [↗](#)). Notably, compared to parental cells, *GMCL1* KO cells, whether reconstituted with WT or mutant GMCL1 constructs, showed no significant alterations in cell cycle profiles (**Supplementary Figure 2D** [↗](#)).

Next, we used cycloheximide (CHX) to inhibit translation and to assess the stability of 53BP1. We observed that in the chromatin-bound fraction of *GMCL1* KO cells, 53BP1 was more stable compared to cells rescued with GMCL1 WT (**Figure 2C** [↗](#)).

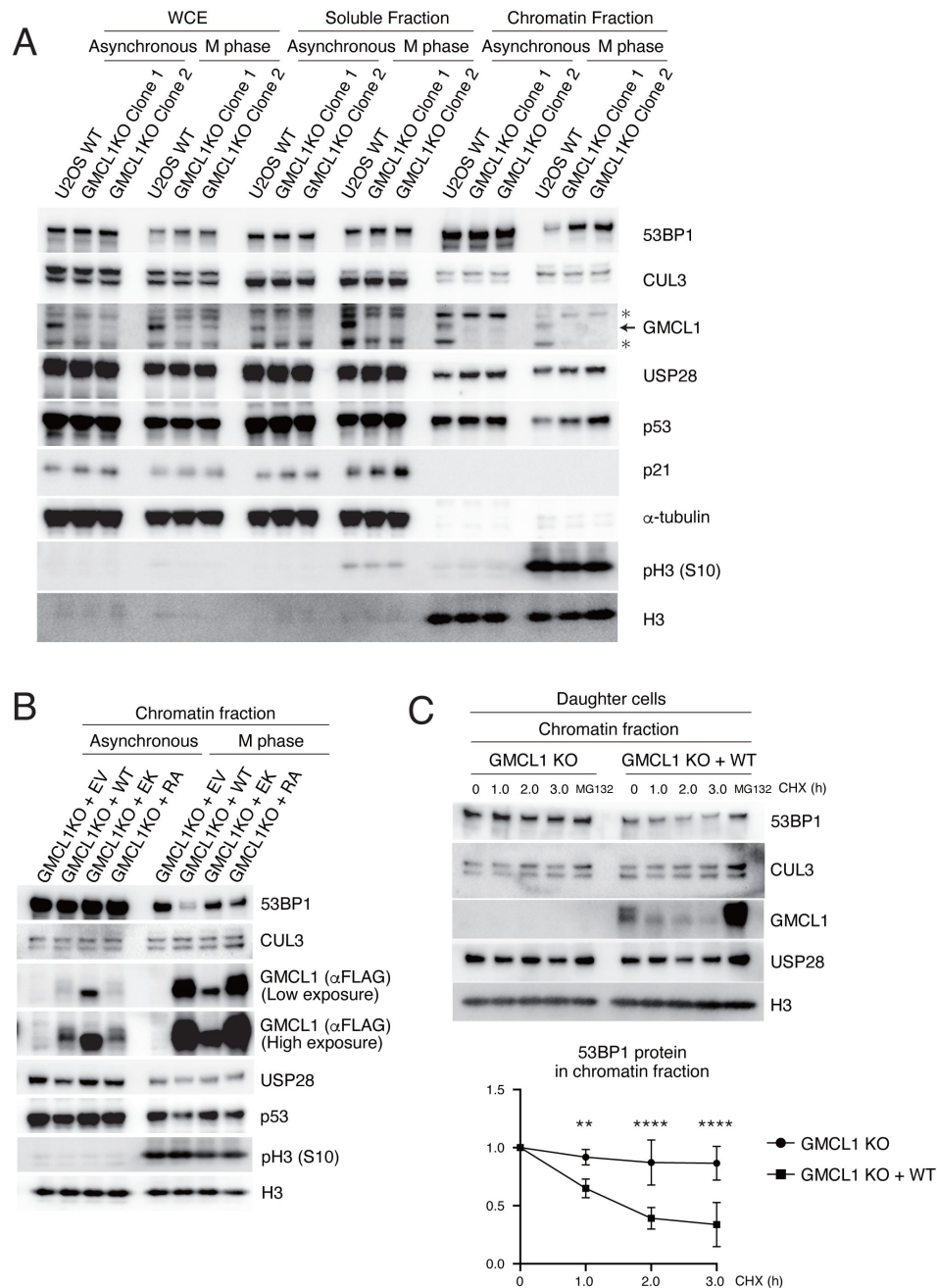


Figure 2.

GMCL1 targets 53BP1 for degradation during M phase

A) Asynchronous or M phase-synchronized WT or GMCL1 knockout (KO) U2OS cells were collected. Whole-cell extracts (WCE) were prepared using RIPA buffer, and other lysates were fractionated into soluble and chromatin-bound fractions for immunoblotting. Arrow indicates GMCL1-specific bands. Asterisk indicates non-specific bands.

B) Stable U2OS cell lines expressing EV, FLAG-GMCL1 WT, FLAG-GMCL1 EK, or FLAG-GMCL1 RA in a GMCL1 KO background were synchronized into M phase and fractionated for chromatin immunoblotting.

C) Mitotic-synchronized FLAG-GMCL1-expressing cells were collected by shake-off and cultured in fresh FBS-containing medium for 7 h. Daughter cells were treated with CHX at the indicated time points and collected within 7 hours, fractionated into soluble and chromatin-bound fractions, and analyzed by immunoblotting.

These results suggest that cells lacking GMCL1 that experienced prolonged mitosis will maintain elevated levels of p53, increasing the likelihood of apoptosis in subsequent cell cycles.

GMCL1 modulates paclitaxel resistance in cancer cells

Paclitaxel, a widely used chemotherapeutic drug, stabilizes microtubules and prevents their depolymerization, thereby halting cancer cell division and inducing apoptosis. To assess the clinical significance of GMCL1, we investigated the relationship between GMCL1 expression levels and Taxol resistance in various cancer cell lines. To this end, we leveraged the PRISM (profiling relative inhibition simultaneously in mixtures) repurposing dataset, which compiles the proliferation inhibitory activity of 4,518 compounds tested across 578 patient-derived cancer cell lines²⁹. This high-throughput screening dataset provides valuable insights into drug resistance mechanisms. We examined the impact of GMCL1 mRNA expression on Taxol resistance by analyzing taxane-related compounds from this dataset, including cabazitaxel, docetaxel, and paclitaxel, integrating it with the Cancer Dependency Map (<https://depmap.org>). Interestingly, we found that several cancer types (*e.g.*: endometrial, breast, and upper aerodigestive tract cancers) with high levels of GMCL1 mRNA exhibited resistance to paclitaxel, cabazitaxel, and/or docetaxel (**Figure 3A-C**). In contrast, lung cancer cell lines with high GMCL1 mRNA expression did not show such tendencies (**Figure 3D**). Given that nearly half of lung cancers harbor p53 mutations³⁰, we hypothesized that p53 status may influence GMCL1-mediated Taxol resistance. To further investigate the functional relationship between GMCL1, 53BP1 and p53 in Taxol resistance, we stratified cancer cells based on their p53 status (wild-type vs mutant) and analyzed GMCL1 mRNA expression and 53BP1 protein levels. In cancer cells with wild type p53, high GMCL1 mRNA expression and low 53BP1 protein levels correlated with significant increased resistance to Cabazitaxel and Paclitaxel when compared to cells with low GMCL1 mRNA and high 53BP1 protein levels (**Figure 3E**). However, this effect was abolished in p53-mutant cells, where GMCL1 status no correlated with Taxol resistance (**Figure 3F**).

To further verify the impact of GMCL1 levels on Taxol sensitivity, we performed cell viability and apoptosis assays using cells with wild type or mutant p53. In p53 wild-type cells (MCF7 and U-2OS), Taxol treatment led to a significant reduction in cell viability and an increase in apoptosis in GMCL1-depleted cells compared to cells transfected with non-targeting control siRNA (**Figure 4A-D**). However, GMCL1 knockdown did not affect cell viability or apoptosis in Taxol-treated cells with mutant p53 or inactivated p53 (HeLa and HEC-1-A, respectively) (**Figure 4E-H**).

These observations are consistent with the results shown in **Figure 3** and support the hypothesis that Taxol resistance is mediated by GMCL1 through the USP28-p53-53BP1 complex. Specifically, high GMCL1 expression promotes the degradation of 53BP1, maintaining low p53 levels, thereby reducing Taxol-induced cell death in cell that have functional p53.

Discussion

We identify GMCL1, a previously uncharacterized human CRL3 substrate receptor, as a regulator of taxane resistance. Specifically, we found that GMCL1 interacts with and mediates the degradation of 53BP1 during prolonged arrest in M phase. While 53BP1 is well known for its role in double-strand break (DSB) repair via non-homologous end joining (NHEJ)³¹, it also participates to a so-called mitotic stopwatch composed of the 53BP1-USP28-p53 complex that stabilizes p53 during prolonged mitotic arrest^{7,9}. We showed that GMCL1 controls the levels of 53BP1 and, consequently, those of p53 in mitotic cells, thereby influencing p53 transmission to daughter cells (see model in **Figure 4I**). This data suggests that GMCL1 functions as a mitotic stress regulator, with potential oncogenic properties in certain contexts.

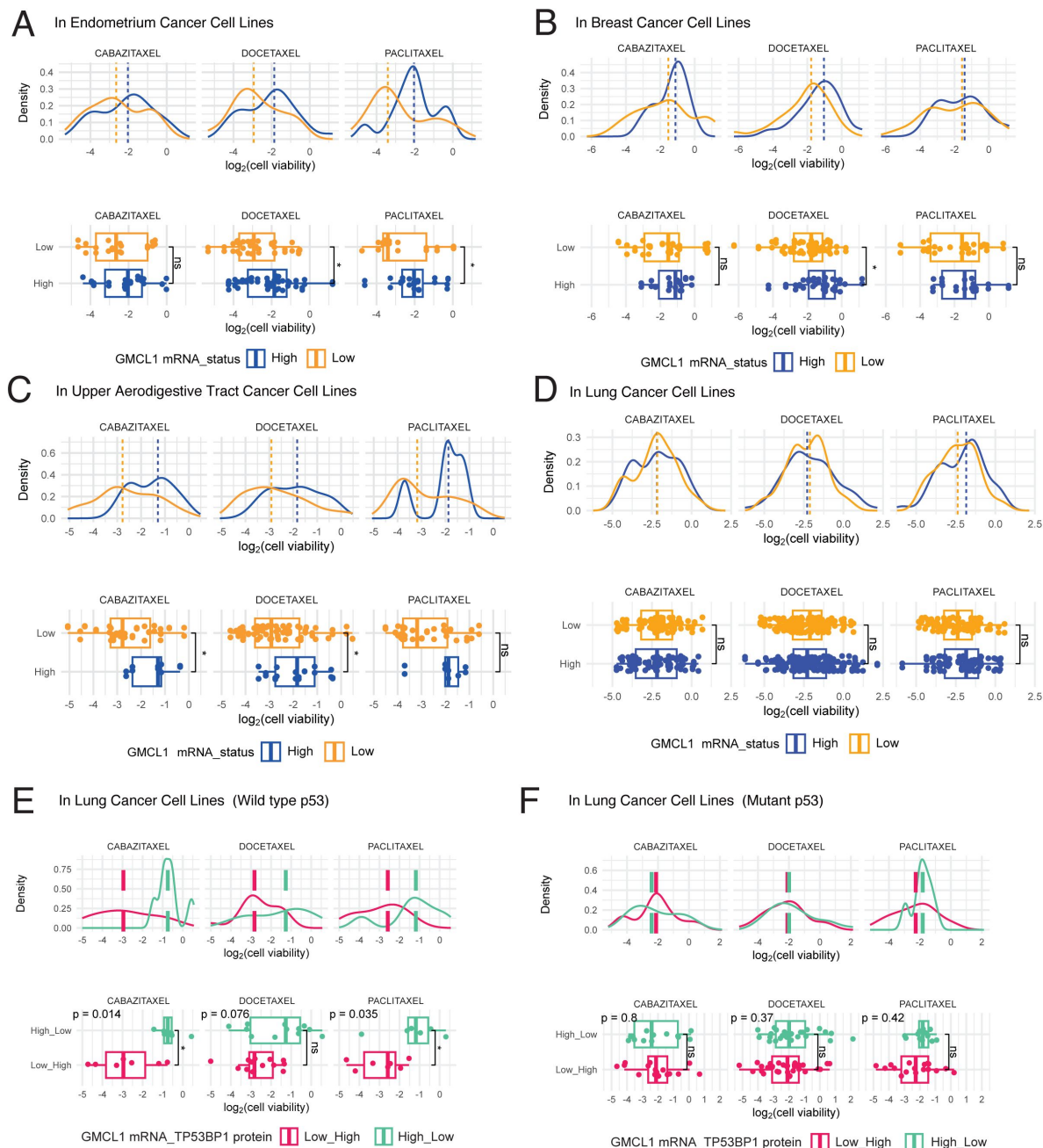


Figure 3.

GMCL1 expression shows positive correlation with Taxol resistance in cancer cell lines

For each cancer type (A-D) and p53 status (E-F), two visualization methods are presented: Upper panels: Density plots show the distribution of drug responses ($\log_2(\text{cell viability})$) to the indicated taxane. Each curve represents the frequency of cell lines exhibiting specific response values. Blue curves represent high GMCL1 expression, while orange curves show low GMCL1 expression. Vertical dashed lines indicate median response values for each group. Rightward shifts of blue curves (high GMCL1) indicate greater resistance.

Lower panels: Boxplots of the same data showing median (horizontal line), interquartile range (box), and distribution range (whiskers). Higher values on the y-axis indicate greater resistance to the drug. Statistical significance is indicated (* $p < 0.05$; ns: not significant) using Wilcoxon rank-sum test.

In panels E-F, cell lines are stratified by both GMCL1 mRNA and 53BP1 protein levels as 'Low_High' (low GMCL1/high 53BP1, red) or 'High_Low' (high GMCL1/low 53BP1, green). Note the significant differences in panels E (wild-type p53) but not in panels F (mutant p53), demonstrating that GMCL1-mediated taxane resistance requires functional p53 signaling.

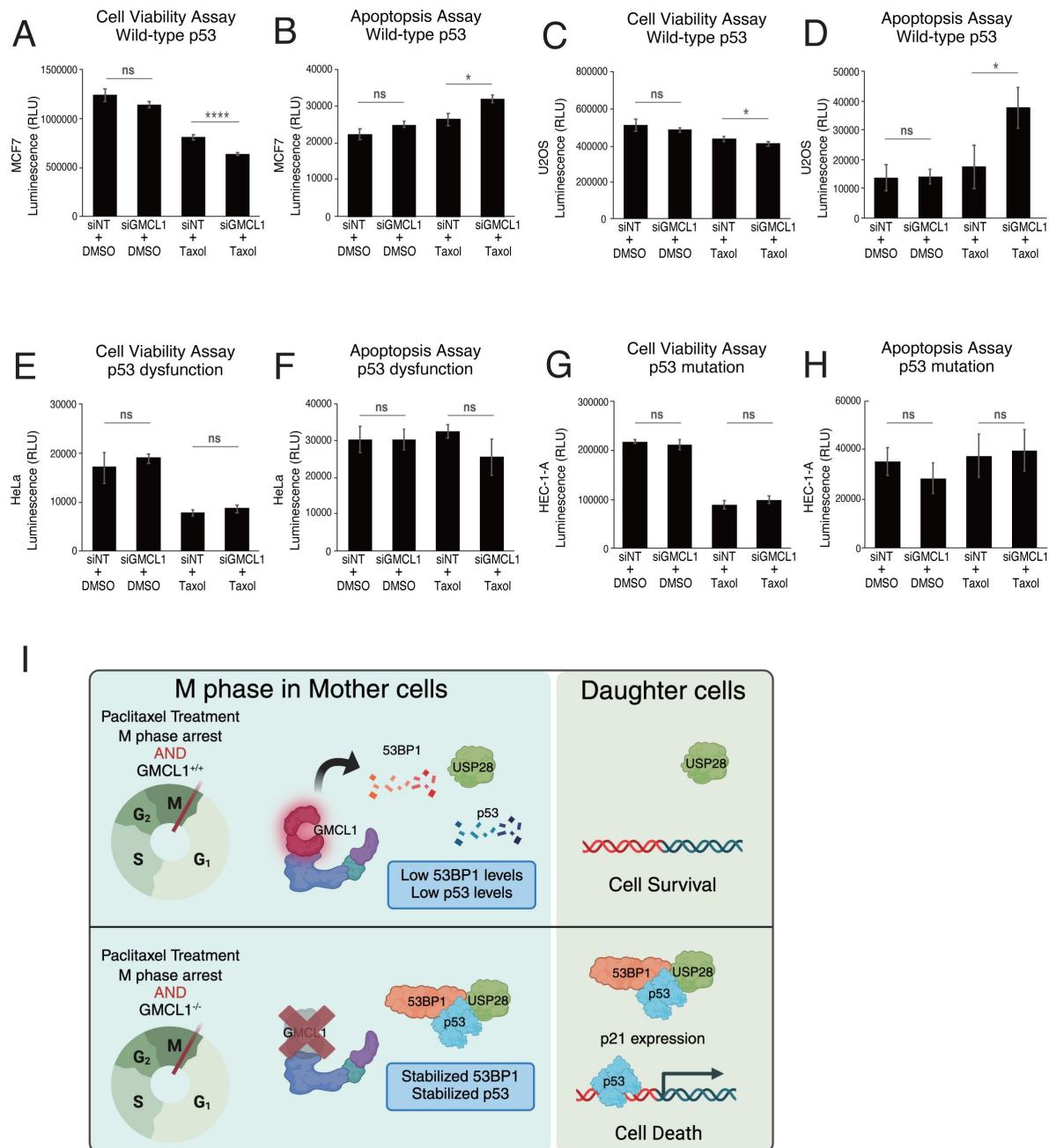


Figure 4.

GMCL1 deficiency sensitizes cancers with wild-type p53 to Taxol-induced apoptosis

MCF7 (A), U2OS (C), HeLa (E), HEC-1-A (G) cells were transfected with GMCL1-targeting siRNAs or non-targeting (NT) control for 72 h. Cells were treated with DMSO or Taxol for 48 h, and cell viability was assessed using the CellTiter-Glo Cell Viability Assay. Error bars represent standard deviation. Apoptosis was measured in the same conditions, i.e., MCF7 (B), U2OS (D), HeLa (F), HEC-1-A (H), using the RealTime-Glo Annexin V Apoptosis and Necrosis Assay. Error bars represent standard deviation.

(I) Overview of GMCL1's function during the M phase. In cancers with high GMCL1 levels, the CRL3^{GMCL1}-mediated degradation of 53BP1 prevents the formation of the mitotic stopwatch complex, leading to p53 degradation and sustained proliferation. Loss of GMCL1 stabilizes mitotic 53BP1 levels and restores paclitaxel sensitivity.

Cancer database analysis reveals that GMCL1 is overexpressed in multiple cancers, including bladder, brain, breast, kidney, liver, lung, ovary, prostate, skin, and stomach cancers (**Supplementary Figure 3** [↗](#)). This underscores GMCL1's role in mitotic regulation and genomic stability, which may vary based on tumor type, genetic background, and additional oncogenic mutations. Recent studies indicate that the MDM2-p53 axis¹⁰ [↗](#), along with the USP28-53BP1-p53 complex, functions as a mitotic timer enforcing G1 arrest following prolonged cell division. Our study identifies GMCL1 as an E3 ligase regulating this pathway by mediating 53BP1 degradation in mitosis. As often is the case for crucial regulatory proteins, 53BP1 is subject to regulation by multiple factors under different cellular circumstances³² [↗](#)–³⁴ [↗](#).

GMCL1 has been primarily studied in the germ cells of *D. melanogaster*, where we have shown that it forms an active CRL3 complex during mitosis. Our new findings establish a critical role for GMCL1 in somatic cell fate decisions by regulating 53BP1 stability during prolonged mitosis. We found that GMCL1 loss enhances Taxol-induced cell death in p53 wild-type cancers but has no effect in p53-mutant cells (**Figure 4I** [↗](#)). This suggests that GMCL1 inhibition could be selectively effective in cancers that retain functional p53. These findings position GMCL1 as a key modulator of mitotic stress and Taxol resistance, highlighting its potential as a therapeutic target in cancer treatment.

Materials and methods

Cell culture

Cell lines were purchased from ATCC and were routinely checked for mycoplasma contamination with the MycoStrip Mycoplasma Detection Kit (Invivogen). HEK293T (ATCC CRL-3216), HeLa (ATCC CCL-2) were maintained in Dulbecco's modified Eagle's medium (DMEM) (Gibco). U-2 OS (ATCC HTB-96), HCT-116 (ATCC CCL-247) and HEC-1-A (ATCC HTB-112.NM) cells were maintained in McCoy's 5A medium (Gibco). MCF7 (ATCC HTB-22) were maintained in Eagle's Minimum Essential Medium (EMEM) (ATCC). All media were supplemented with 10% fetal bovine serum (FBS) (Corning Life Sciences) and 1% penicillin/streptomycin/L-glutamine (Corning Life Sciences); however, MCF7 was further supplemented with 0.28% human recombinant insulin, zinc solution (11.2 µg/ml, Gibco). All cell lines were maintained at 37 °C and 5% CO₂ in a humidified atmosphere.

Plasmids, siRNA, and transfection

Homo sapiens cDNAs were amplified by PCR using KAPA HiFi DNA Polymerase (Kapa Biosystems) and sub-cloned into a variety of vector backbones, including modified pCDNA3.1 vectors containing C-terminal Flag tags, and pLVX-PURO lentiviral vectors containing C-terminal Flag. Site-directed mutagenesis was performed using KAPA HiFi DNA Polymerase (Kapa Biosystems). All cell lines were transiently transfected using Lipofectamine 3000 (ThermoFisher Scientific) based on the manufacturer's recommendation. siRNA oligo transfections were performed using RNAiMax (ThermoFisher Scientific) according to the manufacturer's instructions.

Virus-mediated gene transfer

For the generation of lentivirus, HEK293T cells were transfected with pLVX constructs carrying the genes of interest, alongside the packaging plasmids pCMV-delta-R8.2 and pCMV-VSV-G. Viral supernatant was harvested 48 hours post-transfection, passed through a 0.45-µm sterile Millex-HV filter unit (Millipore Sigma), and supplemented with polybrene at a final concentration of 8 µg/ml (Sigma). Target cells were infected by replacing their culture medium with the virus-containing supernatant for an 8-hour incubation period. Selection of successfully transduced cells was performed using puromycin at a concentration of 1–2 µg/ml (Sigma).

CRISPR-Cas9 genome editing

CRISPR-Cas9 genome editing techniques were carried out as previously described³⁵ with modifications. In brief, to generate *GMCL1*-knockout U-2 OS cells, optimal gRNA target sequences closest to the start codon of the genes were designed using the Benchling CRISPR Genome Engineering tool (<https://www.benchling.com>). For transient Cas9 expression, gRNAs specific for *GMCL1* gene was incorporated into the pRP [CRISPR]-Hygro-hCas9-U6 vector, which was obtained from VectorBuilder (<https://en.vectorbuilder.com/>). The following oligos were used to generate the proper gRNA in the vector: *GMCL1* (5'-CGTGCCCCACGTACCTTCG-3'). To generate *GMCL1* 2×Flag knock-in HCT116 cells, an optimal gRNA target sequence closest to the genomic target site and a ~2 kb homologous recombination (HR) donor template was designed using the Benchling CRISPR Genome Engineering tool. The HR donor template was designed to introduce a 2×Flag tag in frame with the C terminus of *GMCL1*, in the following order: *GMCL1*-linker-FLAG-linker-FLAG-Stop codon and was purchased from VectorBuilder (<https://en.vectorbuilder.com/>). The following single gRNA sequence was used for the transient hCas9 expression vector: *GMCL1* (5'-AAGTTACAGCAGATATATAA-3').

Genomic DNA was collected using QuickExtract (Epicentre). Genotyping PCRs were performed with MyTaq HS Red Mix (Bioline), using primers surrounding the genomic target sites. The following primers were used for genotyping: *GMCL1* (F: 5'-GCAGGCTTCTGATCTCCCT-3', R: 5'-ACTTGTCATCGTCGTCCTTGT-3'), and *GMCL1* (F: 5'-GGGTGGGAGTTTGGAGAGTG-3', R: 5'-TCTGGATTTCTGGGTGACGA-3'). The resulting PCR products were then purified and sequenced to determine the presence of insertion or deletion events. Clones positive to insertion or deletion events were then validated by western blot.

Antibodies

The following antibodies were used: β -actin (1:5,000, Sigma-Aldrich A5441), CUL3 (1:1,000, Bethyl Laboratories A301-109A), FLAG (1:2,000, Sigma-Aldrich F7425), *GMCL1* (1:1,000, Proteintech 15575-1-AP), HA (1:2,000, Bethyl Laboratories A190-108A), Histone H3 (1:1,000, Abcam, ab1791), p21 (1:1,000, Cell Signaling Technology 2947S), p53 (1:1,000, Proteintech 10442-1-AP), pHistone H3 (D2C8) (Ser10, 1:1,000, Cell Signaling Technology, #3377), USP28 (1:1,000, Proteintech 17707-1-AP), 53BP1 (1:2,000, Abcam ab36823), α -tubulin (1:5,000, Sigma-Aldrich T6074).

Drug treatment procedures

Where indicated, cells were treated with 0.4 mg/ml Nocodazole (Sigma-Aldrich M1404) for 16 hours, 10 μ M MG132 for 3 hours, 2.5 μ M MLN4924 for 3 hours, 100 μ g/ml cycloheximide (CHX) for the indicated time.

qRT-PCR

Total RNA was purified using RNeasy mini kits (Qiagen). cDNA was generated using Double Primed EcoDry kits (Takara). The qPCR reaction was carried out using PowerUp SYBR Green (Applied Biosystems) and the Applied Biosystems QuantStudio 3 Real-Time PCR system in a 96-well format. ROX was used as a reference dye for fluorescent signal normalization and for well-to-well optical variations correction. Bar graphs represent the relative ratios of target genes to β -actin housekeeping gene values. For each biological sample, triplicate reactions were analyzed using absolute relative quantification method alongside in-experiment standard curves for each primer set to control for primer efficiency. The oligos used for qRT-PCR analysis were: β -actin (F: 5'-CATGTACGTTGCTATCCAGGC-3', R: 5'-CTCCTTAATGTCACGCACGAT-3'), *GMCL1* (F: 5'-GGAGATTCCTGACCAGAACATTG-3', R: 5'-CGACTGGGCTTTATCAAGACAT-3'), *GMCL2* (F: 5'-CCACGCAGCGGTCTGT, R: 5'-TGGATTTTCTGGGTGACGATTATT), p21 (F: 5'-

TGTCGTCAGAACCCATGC-3', R: 5'-AAAGTCGAAGTTCCATCGCTC-3'), NOXA (F: 5'-CCAAGCCGTGACCAAGGAC-3', R: 5'-CGCCACATTGTGTAGCACCT-3'), PUMA delta (F: 5'-GCCAGATTTGTGAGACAAGAGG-3', R: 5'-CAGGCACCTAATTGGGCTC-3').

Immunoprecipitation, immunoblotting, and fractionation

For whole-cell lysates, cells were directly lysed in a buffer containing 25 mM Tris-HCl (pH 8.0), 150 mM NaCl, 0.2% NP-40, 10% glycerol, 1 mM EDTA, 1 mM EGTA, 2 mM MgCl₂, 1 mM dithiothreitol (DTT), and 5 mM N-ethylmaleimide (Sigma-Aldrich). When cellular fractionation was performed, it followed a previously established method³⁶. Briefly, cells were lysed in CSK buffer (10 mM PIPES, pH 7.0, 150 mM NaCl, 300 mM sucrose, 0.1% Triton X-100, 3 mM MgCl₂, 1 mM EGTA, 1 mM DTT, and 5 mM N-ethylmaleimide) for 5 minutes. The soluble fraction was collected by centrifugation at 3,000 × g for 3 minutes at 4°C. Cell pellets were subsequently washed in CSK buffer and then lysed in chromatin extraction buffer (50 mM Tris-HCl, pH 7.5, 150 mM NaCl, 0.2% NP-40, 10% glycerol, 1 mM EDTA, 1 mM EGTA, 4 mM MgCl₂, 0.5 mM DTT, 5 mM N-ethylmaleimide, and 5 U/mL TurboNuclease (Accelagen)) for 30 minutes. Insoluble debris was removed by centrifugation at 20,000 × g for 15 minutes at 4°C. All buffers were supplemented with protease inhibitors (Complete ULTRA, Roche) and phosphatase inhibitors (PhosSTOP, Roche).

For immunoprecipitation and affinity purification, samples were incubated with FLAG-M2 magnetic beads (Sigma-Aldrich) at 4°C for 2 hours. Beads were thoroughly washed with lysis buffer, and protein elution was performed using either FLAG peptide (Sigma-Aldrich) for mass spectrometry analysis or 1x Laemmli sample buffer for Western Blot analysis.

Immunoblotting

Western blotting was carried out as described previously³⁵. Protein samples were resolved under denaturing and reducing conditions on 4%–12% Bis-Tris gels (NuPAGE) and transferred onto PVDF membranes (Immobilon-P, Millipore). Membranes were blocked with 5% nonfat dried milk, incubated overnight at 4°C with primary antibodies, followed by washes and incubation with HRP-conjugated secondary antibodies (Amersham GE).

Immunoreactive bands were visualized using enhanced chemiluminescence reagents (Pierce) and detected with a ChemiDoc MP imaging system (Bio-Rad). Each Western blot experiment was conducted at least three times to ensure reproducibility, with representative blots shown in the figures.

Cell cycle analysis by flow cytometry

BrdU incorporation and propidium iodide staining were performed following established protocols³⁷. In brief, cells were pulse-labeled with 10 μM BrdU (Merck) for 45 minutes, followed by trypsinization and overnight fixation in ice-cold 70% ethanol. After washing with PBS and subsequently with PBST-BSA (1% BSA, 0.5% Tween-20 in PBS), DNA denaturation was carried out for 30 minutes using 2 N HCl with 0.5% Triton X-100. The cells were then washed in PBS and neutralized by resuspending them in 0.1 M sodium tetraborate (pH 8.5) for 2 minutes. To detect BrdU incorporation, cells were incubated for 1 hour at room temperature with FITC-conjugated anti-BrdU mouse antibodies (BD Transduction Laboratories, 556028) in PBST-BSA. After multiple washes, cells were resuspended in PBST-BSA containing 5 μg/mL propidium iodide (Invitrogen) and 20 μg/mL RNase A (Invitrogen). Flow cytometry analysis of cell cycle distribution was conducted using a CytoFlex Analyzer (Beckman Coulter) and data were processed with FlowJo v10 software (Becton Dickinson).

Mass spectrometry analysis of GMCL1 immunoprecipitations

The eluted anti FLAG-tag antibody purified protein complexes were reduced with 2 μ l of 0.2 M DTT for 1 h at 57 °C and subsequently alkylated with 2 μ l of 0.5 M iodoacetamide (Sigma) for 45 min at room temperature in the dark. 250 ng of SP3 beads (Cytiva) were added proteins precipitated onto the beads by adding ethanol. Samples were placed in a thermomixer at 25°C for 10 min. Beads were washed three times with 80% ethanol and then digested overnight with 400ng of sequencing grade modified Trypsin (Promega) in 100 mM ammonium bicarbonate. Next, the samples were spun down 21,000 x g for 1 minute. The supernatant was transferred to a new tube while the beads were washed twice with 0.5% acetic acid. The washes were then combined with the supernatant collection.

Samples were acidified with 10% TFA to pH1 and loaded onto a 0.1% TFA equilibrated Pierce C18 spin column using a microcentrifuge. The samples were rinsed twice with 0.1% TFA and twice more using 0.5% acetic acid. Peptides were eluted with 80% acetonitrile in 0.5% acetic acid. The organic solvent was removed using a SpeedVac concentrator and the sample reconstituted in 0.5% acetic acid.

An equal aliquot of each sample was loaded onto a trap column (Acclaim PepMap 100 pre-column, 75 μ m $\times$ 2 cm, C18, 3 μ m, 100 A, Thermo Scientific) connected to an analytical column (EASY-Spray column, 50 m $\times$ 75 μ m internal diameter, PepMap RSLC C18, 2 μ m, 100 A, Thermo Scientific) using the autosampler of an Easy nLC 1200 (Thermo Fisher Scientific) with solvent A consisting of 2% acetonitrile in 0.5% acetic acid and solvent B consisting of 80% acetonitrile in 0.5% acetic acid. The peptide mixture was gradient eluted using the following gradient: 5% solvent B for 5 minutes, 5–35% solvent B in 60 min, 35–45% solvent B in 10 min, followed by 45–100% solvent B in 10 min. The samples were acquired on the Orbitrap Eclipse using the following parameters: full MS spectra resolution of 120,000, an AGC target of 4e5, maximum ion time of 50 ms, scan range from 400 to 1,500 m/z. The MS/MS spectra were collected with the following parameters: a resolution of 30,000, an AGC target of 2e5, maximum ion time of 30 ms, one microscan, 2 m/z isolation window, normalized collision energy (NCE) of 27 and a dynamic exclusion of 30 s. To identify binding partners, all acquired MS2 spectra were searched against a UniProt human database using Sequest HT within Proteome Discoverer 1.4 (Thermo Fisher Scientific).

Fixed modifications were set on cysteine (carbamidomethyl), variable modifications of oxidation on methionine, and deamidation on glutamine and asparagine. The resulting peptide spectra matches and proteins are filtered to better than 1% false discovery rate (FDR) and only proteins with at least two different peptides are reported. Proteins differentially expressed between GMCL1 WT and EK, as determined by SAINT scores with a 5% FDR, were considered significantly enriched interactions when comparing to GMCL1 BTB.

Taxol resistance analysis

Taxol resistance was analyzed across cancer cell lines catalogued in DepMap 24Q2. We integrated TP53 mutation status, *GMCL1* RNA and *TP53BP1* protein expression information across cells with the PRISM dataset²⁹ [\[link\]](#), to evaluate taxol resistance in the context of *GMCL1* or *TP53BP1* levels and *TP53* mutation.

Cell viability and apoptosis assays

Cells (2500/well) were plated in a 96-well plate. The medium was replaced with 50 μ L of medium containing the target siRNA (0.2 μ L/well). After 24 hours, an additional 50 μ L of the designated reagent was added. Luminescence was measured using BioTek Synergy Neo2 (Agilent) 48 hours post-treatment with CellTiter-Glo 2.0 Cell Viability Assay (Promega) or RealTime-Glo Annexin V Apoptosis and Necrosis Assay (Promega) according to the manufacturer's protocol.

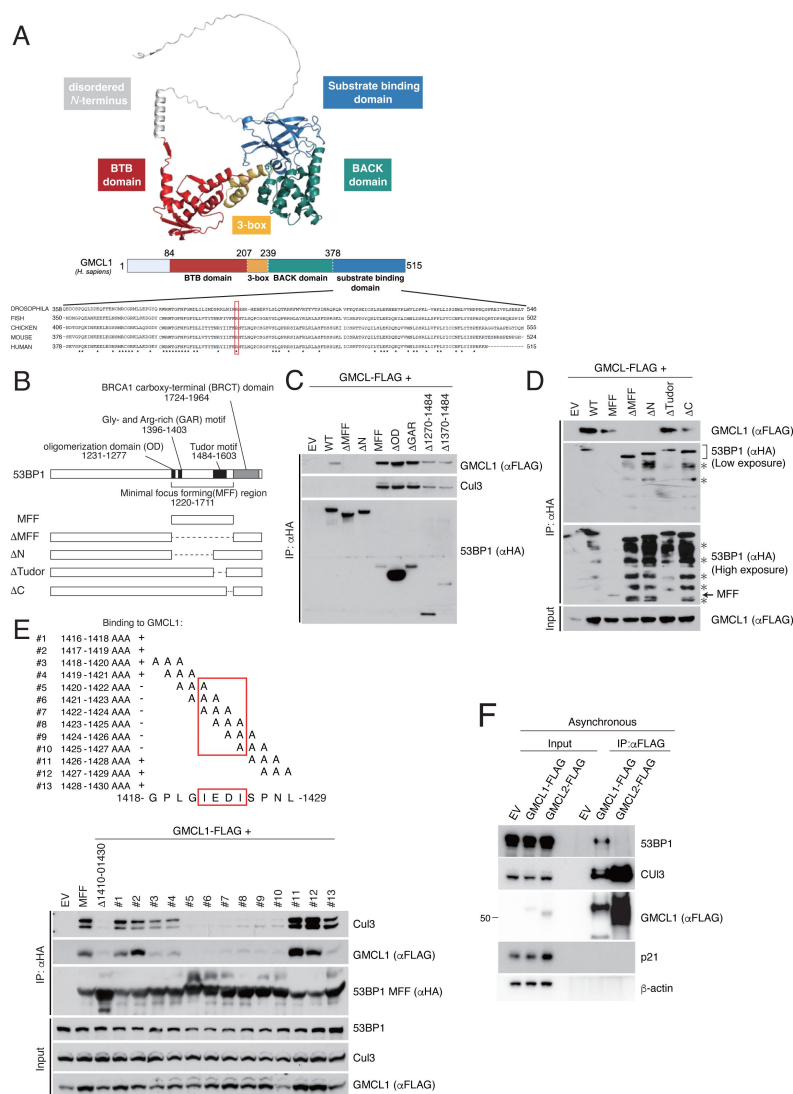
Quantification and statistical analysis

Data analysis was performed using GraphPad Prism version 10.2.1. For comparisons involving three or more groups, One-way ANOVA followed by Bonferroni's post hoc test or the Brown-Forsythe and Welch ANOVA test followed by Dunnett's T3 multiple comparisons test was applied. For bioinformatic data analysis, two-group comparisons were performed using the pairwise Wilcoxon rank-sum test.

Data and code availability

- Original western blot images have been deposited at Mendeley at DOI:10.17632/gj3x6r263d.1 and are publicly available as of the date of publication. <https://data.mendeley.com/preview/gj3x6r263d?a=1e452dfc-bf85-472c-83ce-0e997ba6fa40>
- The mass spectrometric raw files are accessible at <https://massive.ucsd.edu> under accession MassIVE MSV000097235 and at www.proteomexchange.org under accession PXD061458. Username: MSV000097235_reviewer, Password: TP53bdg-protein1

Supplemental figures



Supplemental Figure 1.

Mapping 53BP1 binding sites on GMCL1

(A) Predicted structure of GMCL1, domain architecture overview, and comparative analysis of the substrate-binding domain across *Drosophila*, fish, chicken, mouse, and human. Conserved amino acids are indicated by asterisks, with the human R433 residue highlighted in red.

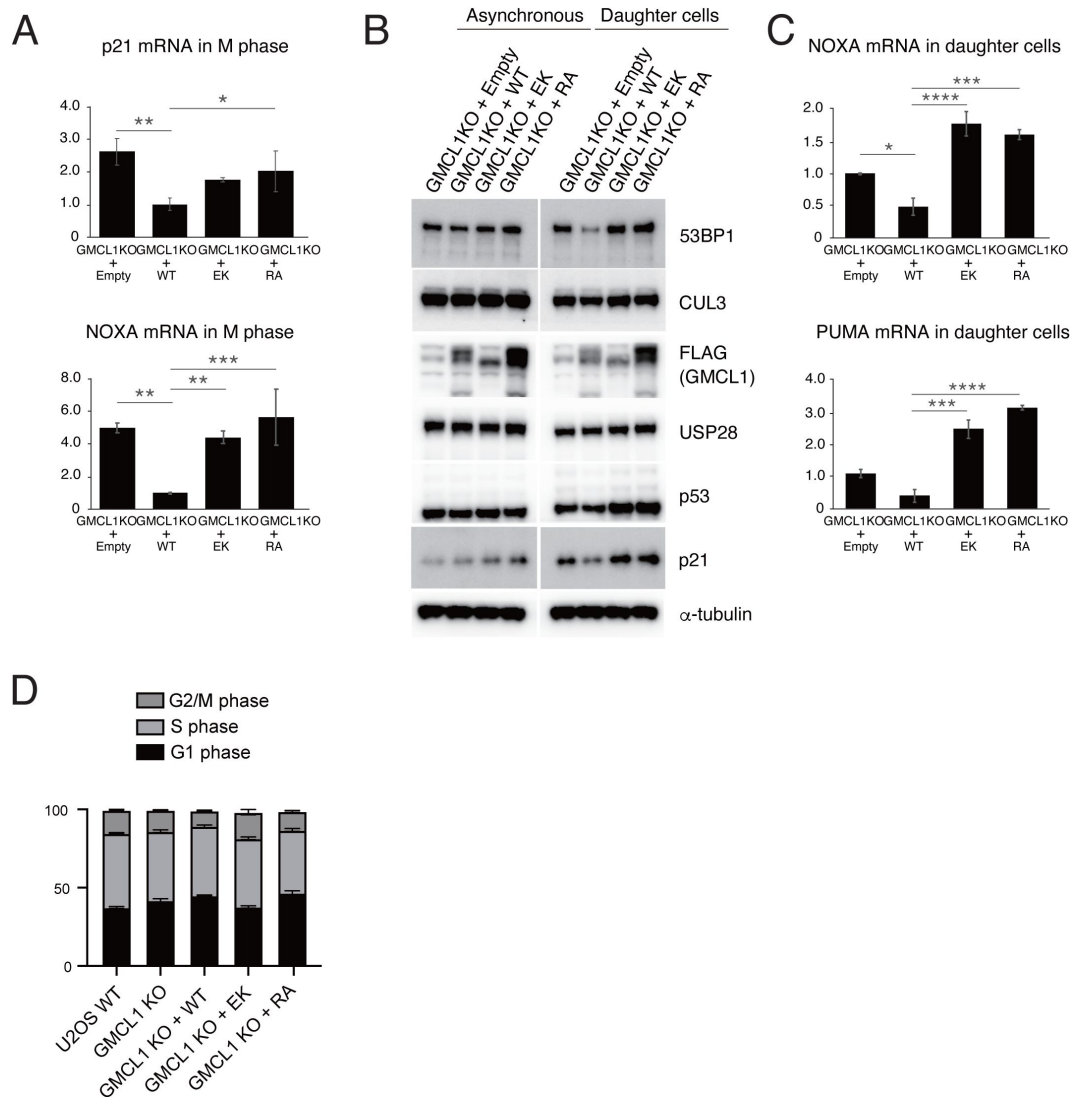
(B) Schematic representation of 53BP1 domains.

(C) HEK293T cells were co-transfected with FLAG-GMCL1 and either EV, HA-53BP1 WT, or deletion mutants: HA-53BP1 ΔMFF, HA-53BP1 ΔN (lacking the N-terminus of MFF), HA-MFF, HA-MFF ΔOD (oligomerization domain), HA-MFF ΔGAR (glycine-arginine-rich motif), HA-MFF Δ1270-1484, or HA-MFF Δ1370-1484. Immunoprecipitation of 53BP1 was performed using HA beads, followed by immunoblotting of co-purified proteins.

(D) HEK293T cells were transfected with EV or FLAG-GMCL1, together with HA-53BP1 WT or mutants: HA-MFF, HA-53BP1 ΔMFF, HA-53BP1 ΔN (lacking the N-terminus of MFF), HA-53BP1 ΔTudor, and HA-53BP1 ΔC (lacking the C-terminus of MFF). 53BP1 was immunoprecipitated with HA-beads, followed by immunoblotting of co-purified proteins. Asterisk indicates non-specific bands.

(E) To narrow down the GMCL1-binding region on 53BP1, sequential 20-amino-acid deletions within the MFF domain were generated. HEK293T cells were co-transfected with EV or FLAG-GMCL1, along with HA-MFF, HA-MFF Δ1410-1430, and site-specific mutants (every three amino acids mutated within 1410–1430 region of 53BP1). Immunoprecipitation of 53BP1 was conducted with HA-beads, followed by immunoblotting.

(F) HEK293T cells were transfected with FLAG-GMCL1 or FLAG-GMCL2. GMCL1 and GMCL2 were immunoprecipitated with FLAG-beads and analyzed by immunoblotting.



Supplemental Figure 2.

Mitotic stress imprints apoptotic memory in daughter cells

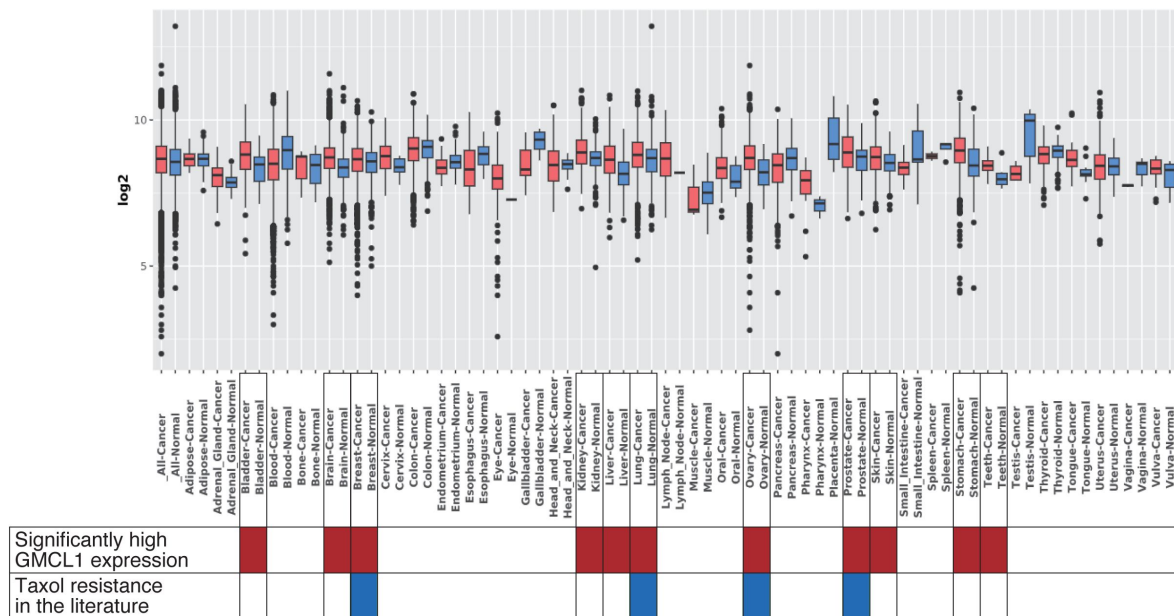
(A) RNA was extracted from FLAG-GMCL1-expressing U2OS cells (as in [Figure 2B](#)). p21 and NOXA mRNA levels were quantified by qPCR from three independent experiments. Error bars represent standard deviation.

(B) Stable GMCL1 KO U2OS cells were transduced with lentiviruses expressing EV, FLAG-GMCL1 WT, FLAG-GMCL1 EK, or FLAG-GMCL1 RA. Cells were synchronized in M phase, then released into fresh FBS-containing medium for 20 hours following shake-off. Daughter cells were fractionated into soluble and chromatin-bound fractions and analyzed by immunoblotting.

(C) RNA was extracted from same cells as in (B), and NOXA and PUMA mRNA levels were quantified by qPCR from three independent experiments. Error bars represent standard deviation.

(D) Cell cycle distribution was determined using BrdU pulse and PI staining of asynchronous U2OS WT or FLAG-GMCL1-expressing cells. Error bars represent standard deviation.

GENT2 - Boxplot for GMCL1 gene expression profile across cancer experiments



Supplemental Figure 3.

Tissue-specific comparison of GMCL1 gene expression levels

The panel shows GMCL1 expression levels in normal cells and cancer cells based on GENT2²¹. The cells in the bottom table indicate where, GMCL1 expression levels were significantly higher in cancer cells (red), while the blue cells indicate cancers that were previously reported in the literature to exhibit Taxol resistance^{13,38}.

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Additional information

Author contributions

Y.K., G.R. and M.P. conceived, supervised, and coordinated the study. T.G.R. performed all analyses from PRISM and DepMap data. Y.K. performed the mass spectrometric sample preparation, generated CRISPR KO and KI cell lines, designed plasmid constructs, and performed the cell-based immunoprecipitations, fractionation and protein stability assays. A.M. and J.P. performed the initial mass spectrometric sample preparation and immunoprecipitations. B.M.U and J.O.P performed mass spectrometry analysis. S.F.G. performed structural predictions using AlphaFold3. The manuscript was written with inputs from all authors.

Additional files

Supplementary Table 1 [↗](#)

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Reviewer #1 (Public review):

In this manuscript, Pagano and colleagues test the idea that the protein GMCL1 functions as a substrate receptor for a Cullin RING 3 E3 ubiquitin ligase (CUL3) complex. Using a pulldown approach, they identify GMCL1 binding proteins, including the DNA damage scaffolding protein 53BP1. They then focus on the idea that GMCL1 recruits 53BP1 for CUL3-dependent ubiquitination, triggering subsequent proteasomal degradation of ubiquitinated 53BP1.

In addition to its DNA damage signalling function, in mitosis, 53BP1 is reported to form a stopwatch complex with the deubiquitinating enzyme USP28 and the transcription factor p53 (PMID: 38547292). These 53BP1-stopwatch complexes generated in mitosis are inherited by G1 daughter cells and help promote p53-dependent cell cycle arrest independent from DNA damage (PMID: 38547292). Several studies show that knockout of 53BP1 overcomes G1 cell cycle arrest after mitotic delays caused by anti-mitotic drugs or centrosome ablation (PMID: 27432897, 27432896). In this model, it is crucial that 53BP1 remains stable in mitosis and more stopwatch complex is formed after delayed mitosis.

Pagano and coworkers suggest that 53BP1 levels can sometimes be suppressed in mitosis if the cells overexpress GMCL1. They carry out a bioinformatic analysis of available public data for p53 wild-type cancer cell lines resistant to the anti-mitotic drug paclitaxel and related compounds. Stratifying GMCL1 into low and high expression groups reveals a weak ($p = 0.05$ or ns) correlation with sensitivity to taxanes. It is unclear on what basis the authors claim paclitaxel-resistant and p53 wild-type cancer cell lines bypass the mitotic surveillance/timer pathway. They have not tested this. Figure 3 is a correlation assembled from public databases but has no experimental tests. Figure 4 looks at proliferation but not cell cycle progression or the length of mitosis. The main conclusions relating to cell cycle progression and specifically the link to mitotic delays are therefore not supported by experimental data. There is no imaging of the cell cycle or cell fate after mitotic delays, or analysis of where the cells arrest in the cell cycle. Most of the cell lines used have been reported to lack a functional mitotic surveillance pathway in the recent work by Meitinger. To support these conclusions, the stability of endogenous 53BP1 under different conditions in cells known to have a functional mitotic surveillance pathway needs to be examined. A key suggestion in the work is that the level of GMCL1 expression correlates with resistance to taxanes. For the mitotic surveillance pathway, the type of drug (nocodazole, taxol, etc) used to induce a delay isn't thought to be relevant, only the length of the delay. Do GMCL1-overexpressing cells show resistance to anti-mitotics in general?

Importantly, if GMCL1 specifically degrades 53BP1 during prolonged mitotic arrests, the authors should show what happens during normal cell divisions without any delays or drug treatments. How much 53BP1 is destroyed in mitosis under those conditions? Does 53BP1 destruction depend on the length of mitosis, drug treatment, or does 53BP1 get degraded every mitosis regardless of length? Testing the contribution of key mitotic E3 ligase activities on mitotic 53BP1 stability, such as the anaphase-promoting complex/cyclosome (APC/C) is important in this regard. One previous study reported an analysis of putative APC/C KEN-box

degron motifs in 53BP1 and concluded these play a role in 53BP1 stability in anaphase (PMID: 28228263).

There is no direct test of the proposed mechanism, and it is therefore unclear if 53BP1 is ubiquitinated by a GMCL1-CUL3 ligase in cells, and how efficient this process would be at different cell cycle stages. A key issue is the lack of experimental data explaining why the proposed mechanism would be restricted to mitosis. Indirect effects, such as loss of 53BP1 from the chromatin fraction during M phase upon GMCL1 overexpression, do not necessarily mean that 53BP1 is degraded. PLK1-dependent chromatin-cytoplasmic shuttling of 53BP1 during mitotic delays has been described previously (PMID: 38547292, 37888778). These papers are cited in the text, but the main conclusions of those papers on 53BP1 incorporation into a stopwatch complex during mitotic delays have been ignored. Are the authors sure that 53BP1 is destroyed in mitosis and not simply re-localised between chromatin and non-chromatin fractions? At the very least, these reported findings should be discussed in the text.

The authors use a variety of cancer cell line models throughout their study, most of which have been reported to lack a functional mitotic surveillance pathway. U2OS and HCT116 cells do not respond normally to mitotic delays, despite being annotated as p53 WT. Other studies have used p53 wild-type hTERT RPE-1 cells to study the mitotic surveillance pathway. If the model is correct, then over-expressing GMCL1 in hTERT-RPE1 cells should suppress cell cycle arrest after mitotic delays, and GMCL1 KO should make the cells more sensitive to delays. These experiments are needed to provide an adequate test of the proposed model.

To conclude, while the authors propose a potentially interesting model on how GMCL1 overexpression could regulate 53BP1 stability to limit p53-dependent cell cycle arrest, it is unclear what triggers this pathway or when it is relevant. 53BP1 is known to function in DNA damage signalling, and GMCL1 might be relevant in that context. The manuscript contains the initial description of GMCL1-53BP1 interaction but lacks a proper analysis of the function of this interaction and is therefore a preliminary report.

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Reviewer #2 (Public review):

Summary:

This study investigates the role of GMCL1 in regulating the mitotic surveillance pathway (MSP), a protective mechanism that activates p53 following prolonged mitosis. The authors identify a physical interaction between 53BP1 and GMCL1, but not with GMCL2. They propose that the ubiquitin ligase complex CUL3-GMCL1 targets 53BP1 for degradation during mitosis, thereby preventing the formation of the "mitotic stopwatch" complex (53BP1-USP28-p53) and subsequent p53 activation. The authors show that high GMCL1 expression correlates with resistance to paclitaxel in cancer cell lines that express wild-type p53. Importantly, loss of GMCL1 restores paclitaxel sensitivity in these cells, but not in p53-deficient lines. They propose that GMCL1 overexpression enables cancer cells to bypass MSP-mediated p53 activation, promoting survival despite mitotic stress. Targeting GMCL1 may thus represent a therapeutic strategy to re-sensitize resistant tumors to taxane-based chemotherapy.

Strengths:

This manuscript presents potentially interesting observations. The major strength of this article is the identification of GMCL1 as a 53BP1 interaction partner. The authors identified relevant domains and showed that GMCL1 controls 53BP1 stability. The authors further show a potentially interesting link between GMCL1 status and sensitivity to Taxol.

Weaknesses:

However, the manuscript is significantly weakened by unsubstantiated mechanistic claims, overreliance on a non-functional model system (U2OS), and overinterpretation of correlative data. To support the conclusions of the manuscript, the authors must show that the GMCL1-dependent sensitivity to Taxol depends on the mitotic surveillance pathway.

<https://doi.org/10.7554/eLife.106730.1.sa1>

Reviewer #3 (Public review):

Summary:

In this study, Kito et al follow up on previous work that identified *Drosophila* GCL as a mitotic substrate recognition subunit of a CUL3-RING ubiquitin ligase (CRL3) complex.

Here they characterize mutants of the human ortholog of GCL, GMCL1, that disrupt the interaction with CUL3 (GMCL1E142K) and that lack the substrate interaction domain (GMCL1 BBO). Immunoprecipitation followed by mass spectrometry identified 9 proteins that interacted with wild-type FLAG-GMCL1 and GMCL1 EK but not GMCL1 BBO. These proteins included 53BP1, which plays a well-characterized role in double-strand break repair but also functions in a USP28-p53-53BP1 "mitotic stopwatch" complex that arrests the cell cycle after a substantially prolonged mitosis. Consistent with the IP-MS results, FLAG-GMCL1 immunoprecipitated 53BP1. Depletion of GMCL1 during mitotic arrest increased protein levels of 53BP1, and this could be rescued by wild-type GMCL1 but not the E142K mutant or a R433A mutant that failed to immunoprecipitate 53BP1.

Using a publicly available dataset, the authors identified a relatively small subset of cell lines with high levels of GMCL1 mRNA that were resistant to the taxanes paclitaxel, cabazitaxel, and docetaxel. This type of analysis is confounded by the fact that paclitaxel and other microtubule poisons accumulate to substantially different levels in various cell lines (DOI: 10.1073/pnas.90.20.9552 , DOI: 10.1091/mbc.10.4.947), so careful follow-up experiments are required to validate results. The correlation between increased GMCL1 mRNA and taxane resistance was not observed in lung cancer cell lines. The authors propose this was because nearly half of lung cancers harbor p53 mutations, and lung cancer cell lines with wild-type but not mutant p53 showed the correlation between increased GMCL1 mRNA and taxane resistance. However, the other cancer cell types in which they report increased GMCL1 expression correlates with taxane sensitivity also have high rates of p53 mutation. Furthermore, p53 status does not predict taxane response in patients (DOI: 10.1002/1097-0142(20000815)89:4<769::aid-cnrc8>3.0.co;2-6 , DOI: 10.1002/(SICI)1097-0142(19960915)78:6<1203::AID-CNCR6>3.0.CO;2-A , PMID: 10955790).

The authors then depleted GMCL1 and reported that it increased apoptosis in two cell lines with wild-type p53 (MCF7 and U2OS) due to activation of the mitotic stopwatch. This is surprising because the mitotic stopwatch paper they cite (DOI: 10.1126/science.add9528) reported that U2OS cells have an inactive stopwatch and that activation of the stopwatch results in cell cycle arrest rather than apoptosis in most cell types, including MCF7. Beyond this, it has recently been shown that the level of taxanes and other microtubule poisons achieved in patient tumors is too low to induce mitotic arrest (DOI: 10.1126/scitranslmed.3007965 , DOI: 10.1126/scitranslmed.abd4811 , DOI: 10.1371/journal.pbio.3002339), raising concerns about the relevance of prolonged mitosis to paclitaxel response in cancer. The findings here demonstrating that GMCL1 mediates degradation of 53BP1 during mitotic arrest are solid and of interest to cell biologists, but it is unclear that these findings are relevant to paclitaxel response in patients.

Strengths:

This study identified 53BP1 as a target of CRL3GMCL1-mediated degradation during mitotic arrest. AlphaFold3 predictions of the binding interface, followed by mutational analysis, identified mutants of each protein (GMCL1 R433A and 53BP1 IEDI1422-1425AAAA) that disrupted their interaction. Knock-in of a FLAG tag into the C-terminus of GMCL1 in HCT116 cells, followed by FLAG immunoprecipitation, confirmed that endogenous GMCL1 interacts with endogenous CUL3 and 53BP1 during mitotic arrest.

Weaknesses:

The clinical relevance of the study is overinterpreted. The authors have not taken relevant data about the clinical mechanism of taxanes into account. Supraphysiologic doses of microtubule poisons cause mitotic arrest and can activate the mitotic stopwatch. However, in physiologic concentrations of clinically useful microtubule poisons, cells proceed through mitosis and divide their chromosomes on mitotic spindles that are at least transiently multipolar. Though these low concentrations may result in a brief mitotic delay, it is substantially shorter than the arrest caused by high concentrations of microtubule poisons, and the one mimicked here by 16 hours of 0.4 mg/mL nocodazole, which is not used clinically and does not induce multipolar spindles. Resistance to mitotic arrest occurs through different mechanisms than resistance to multipolar spindles. No evidence is presented in the current version of the manuscript that GMCL1 affects cellular response to clinically relevant doses of paclitaxel.

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